

Bayesian Reward Filter for Robust RLHF

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3. Methodology

3.1 Bayesian Reward Filter

Let s_t , a_t , s_{t+1} denote the agent's state, action, and next state at timestep t . The environment returns a sparse reward $r_{\text{env},t} \in \{0, 1\}$, received only upon reaching the goal. A human teacher simultaneously provides a scalar feedback signal $r_{\text{human},t} \in [-0.1, 0.1]$.

The total reward delivered to the agent is:

$$r_{\text{total},t} = r_{\text{env},t} + w_t \cdot r_{\text{human},t}$$

where $w_t \in [0, 1]$ is a trust weight computed from a Beta-Bernoulli belief model over the teacher's reliability.

Trust Update Rule.

To assess teacher reliability at each step, the BRF compares the sign of human feedback with the sign of the potential difference $\Delta\phi_t = \phi(s_{t+1}) - \phi(s_t)$, where ϕ is an environment-specific potential function that quantifies objective progress (see Section 3.3). Sign agreement—the teacher encourages the agent when it makes genuine progress and discourages it when it regresses—is counted as evidence of reliability. Disagreement is counted as evidence of unreliability.

A Beta distribution with parameters (α_t, β_t) maintains the belief:

$$\alpha_{t+1} = \alpha_t + 1 \text{ if } \text{sign}(r_{\text{human},t}) = \text{sign}(\Delta\phi_t)$$

$$\beta_{t+1} = \beta_t + 1 \text{ if } \text{sign}(r_{\text{human},t}) \neq \text{sign}(\Delta\phi_t)$$

Steps where $\Delta\phi_t = 0$ are skipped, as they carry no information about teacher reliability. The trust weight is the posterior expected value of the Beta distribution:

$$w_t = \alpha_t / (\alpha_t + \beta_t)$$

The belief state (α_t, β_t) persists across episodes within a training run, allowing trust to accumulate over the full training history.

Prior Initialization.

The filter is initialized with $\alpha_{\blacksquare} = \beta_{\blacksquare} = 1$, corresponding to a Beta(1,1)—the uniform distribution over $[0,1]$. This gives an initial trust weight $w_{\blacksquare} = 0.5$, expressing no prior assumption about teacher reliability. With prior strength $N_{\blacksquare} = \alpha_{\blacksquare} + \beta_{\blacksquare} = 2$, the prior is intentionally weak: even a single observation is sufficient to begin shifting the trust estimate.

3.2 Baselines

The BRF is compared against two baselines representing the boundary cases of the trust weight:

- **Sparse** ($w = 0$): The agent receives only environment reward, ignoring all human feedback. This is the standard RL baseline.
- **Naive** ($w = 1$): The agent fully incorporates all human feedback regardless of reliability, equivalent to standard RLHF with no filtering.

Mode	Trust Weight	Description
Sparse	$w = 0$	Environment reward only
Naive	$w = 1$	Full human feedback incorporated
BRF	$w_t = \alpha_t / (\alpha_t + \beta_t)$	Dynamically updated Bayesian trust

Table 1. Reward mode summary.

3.3 Environments

MiniGrid-Empty-8x8-v0. The agent navigates an open 8×8 room to reach a goal tile. The potential function is $\phi = -\text{manhattan_distance}(\text{agent}, \text{goal})$, which increases monotonically as the agent approaches the goal.

MiniGrid-LavaGapS7-v0. The agent must cross a room containing a lava strip without stepping on it. The potential function penalizes both distance from the goal and proximity to lava: $\phi = -\text{manhattan_distance}(\text{agent}, \text{goal}) - \lambda \cdot \text{lava_adjacency_penalty}$.

MiniGrid-DoorKey-8x8-v0. A three-stage sequential task: collect a key, unlock a door, reach the goal. The potential function is staged—distance to key $\rightarrow$ door $\rightarrow$ goal.

All environments use MiniGrid’s native discrete action space (7 actions) and sparse terminal rewards.

3.4 Synthetic Human Teacher Models

Six synthetic teacher models simulate the range of real-world human feedback quality. All teachers return signals in $[-0.1, 0.1]$, keeping human feedback at 10% of the maximum terminal reward (+1.0).

- **StochasticLow** (noise = 0.1): Correct feedback 90% of the time; flips signal with 10% probability.
- **StochasticHigh** (noise = 0.8): Flips the signal 80% of the time.
- **Adversarial**: Always gives the opposite of correct feedback. Represents a worst-case attacker.
- **DistanceNoise**: Noise probability scales with distance from goal, up to 90% at maximum distance.
- **Fatigue**: Noise increases linearly from 0% to 80% over 50,000 steps.
- **Realistic**: Combines distance confusion, temporal fatigue, and a 5% baseline error rate.

3.5 RL Agents

PPO (Proximal Policy Optimization): An on-policy algorithm that collects fresh rollouts each iteration. The BRF trust weight is applied at collection time, so every gradient update uses transitions

shaped by the current trust estimate.

DQN (Deep Q-Network): An off-policy algorithm that stores past transitions in a replay buffer (size 100,000). Selected to represent off-policy learning and test BRF compatibility with experience replay.

Both agents use an MLP policy (MlpPolicy) over the flattened observation space.

3.6 Hyperparameters

All base RL hyperparameters use Stable-Baselines3 defaults, shared across all reward modes to preserve internal validity.

Parameter	Value
Total timesteps per run	15,000
Epochs (PPO)	10
Learning rate	3×10^{-4}
Optimizer	Adam
Evaluation episodes	20
Random seeds	42, 123, 999
DQN replay buffer size	100,000

Table 2. Base RL hyperparameters (shared across all reward modes).

Parameter	Value	Justification
α (prior agreement count)	1.0	Uninformative uniform prior
β (prior disagreement count)	1.0	Uninformative uniform prior
human_magnitude	0.1	10% of terminal reward

Table 3. BRF-specific hyperparameters.

The full experiment matrix is the Cartesian product of 2 agents $\times$ 3 environments $\times$ 3 reward modes $\times$ 6 human scenarios $\times$ 3 random seeds = **324 training runs**, each evaluated over 20 episodes.

4. Results

4.1 Trust Weight Evolution by Human Type

Figure 1 shows how the Bayesian trust weight w evolves over training for each synthetic human type, under PPO on MiniGrid-Empty-8x8-v0 (mean across three seeds; shaded band shows ± 1 SD). All curves begin at $w = 0.5$ and diverge within the first few hundred steps:

- **Adversarial:** Trust collapses to $w \approx 0.05$ within $\approx 2,000$ steps. Every feedback signal opposes the potential, so nearly every step increments β , driving trust toward zero.

- **StochasticHigh**: Trust stabilizes near $w \approx 0.20$, consistent with a teacher correct only 20% of the time.
- **DistanceNoise**: Trust settles near $w \approx 0.80$. The small 8×8 grid limits maximum distances, so noise never substantially degrades the estimate.
- **Fatigue**: Trust stabilizes near $w \approx 0.90$. The Fatigue teacher’s noise does not peak until 50,000 steps—well beyond the training budget—so no decay is visible. This is expected and discussed in Section 5.3.
- **Realistic**: Trust settles near $w \approx 0.80$, reflecting a noisy but overall reliable teacher.
- **StochasticLow**: Trust rises rapidly to $w \approx 0.90$, correctly reflecting 90% accuracy.

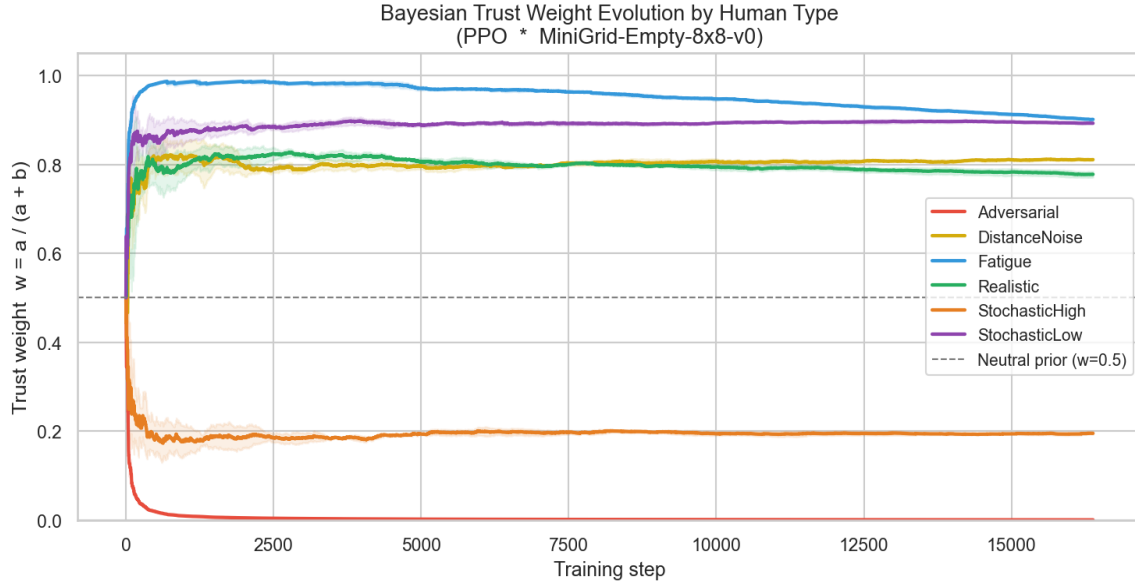


Figure 1. Bayesian trust weight w over training steps by human type (PPO, MiniGrid-Empty-8x8-v0, mean ± 1 SD across 3 seeds). All curves begin at the uniform prior $w = 0.5$.

4.2 PPO Results on MiniGrid-Empty-8x8-v0

Table 4 reports mean success rate across three seeds for PPO on MiniGrid-Empty-8x8-v0; Figure 2 shows the same data as a grouped bar chart.

Human	Sparse	Naive	BRF
StochasticLow	0.000	0.667	0.667
Realistic	0.333	0.667	1.000
Fatigue	0.333	0.333	0.667
DistanceNoise	0.333	0.333	0.333
StochasticHigh	0.000	0.000	0.000
Adversarial	0.000	0.000	0.000

Table 4. PPO success rate by human type and reward mode (MiniGrid-Empty-8x8-v0, mean across 3 seeds).

The BRF achieves its strongest result under Realistic noise: a perfect success rate (1.000) against 0.667 for Naive and 0.333 for Sparse. Under Fatigue, BRF achieves 0.667 compared to 0.333 for both baselines—a 2× improvement. In both cases, BRF is the only condition to improve over Sparse while simultaneously outperforming Naive. Under DistanceNoise all three modes perform equivalently (0.333). StochasticHigh and Adversarial produce 0.000 across all modes.

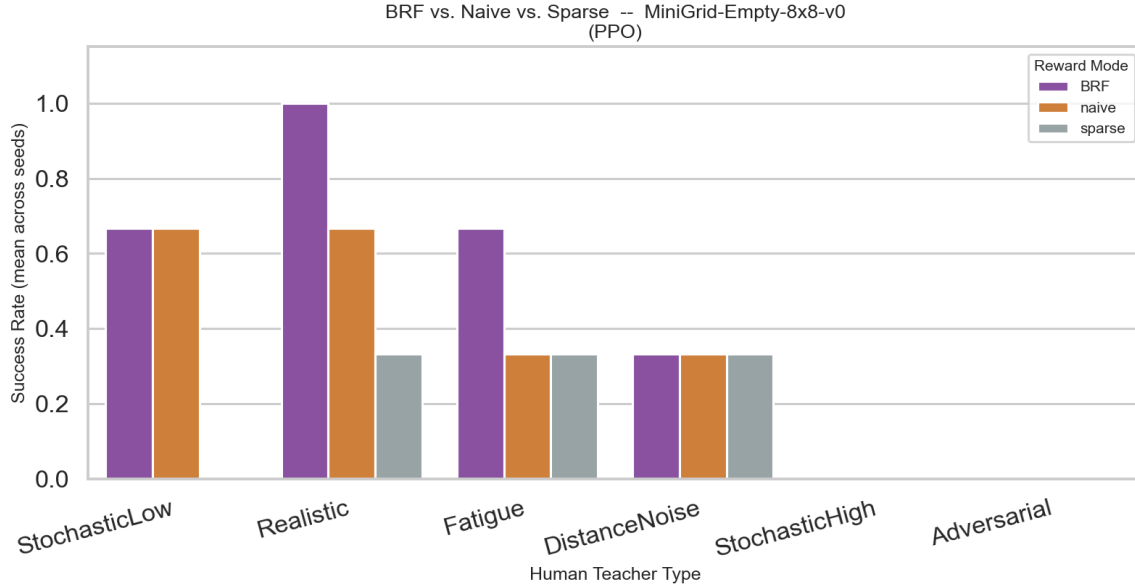


Figure 2. BRF vs. Naive vs. Sparse success rate by human type (PPO, MiniGrid-Empty-8x8-v0, mean across 3 seeds).

4.3 Cross-Environment Performance

MiniGrid-Empty-8x8-v0 is the only environment in which any reward mode achieves meaningful success under PPO. MiniGrid-LavaGapS7-v0 shows minor nonzero success in a small number of human/mode combinations (peak ≈ 0.05 – 0.10 for BRF under Fatigue). MiniGrid-DoorKey-8x8-v0 records 0.000 universally across all modes, humans, and seeds.

4.4 DQN Results

Table 5 reports DQN results across all environments; Figure 3 shows the Empty environment results as a bar chart.

Environment	Human	Sparse	Naive	BRF
DoorKey	All types	0.000	0.000	0.000
Empty	Adversarial	0.000	0.000	0.000
	DistanceNoise	0.000	0.667	0.333
	Fatigue	0.000	0.333	0.000
	Realistic	0.000	0.333	0.333
	StochasticHigh	0.000	0.000	0.000

	StochasticLow	0.000	0.333	0.667
LavaGap	Fatigue	0.000	0.083	0.200
	Realistic	0.000	0.183	0.067
	Others	0.000	≈ 0	0–0.1

Table 5. DQN success rate by environment, human type, and reward mode (mean across 3 seeds).

DQN’s sparse baseline is 0.000 in every condition—unlike PPO, DQN cannot learn from environment reward alone within 15,000 steps. BRF outperforms Naive for StochasticLow on Empty (0.667 vs. 0.333) but underperforms Naive for DistanceNoise and Fatigue.

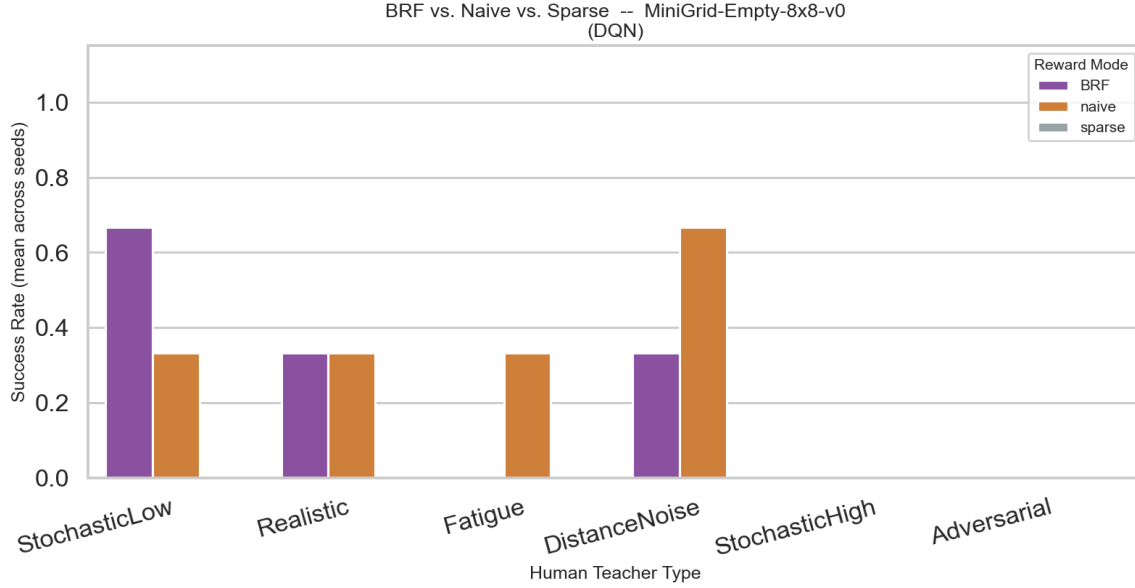


Figure 3. BRF vs. Naive vs. Sparse success rate by human type (DQN, MiniGrid-Empty-8x8-v0, mean across 3 seeds).

5. Analysis and Discussion

5.1 Where BRF Succeeded

On MiniGrid-Empty-8x8-v0 with PPO, the BRF matched or exceeded both baselines in four of six human scenarios. The two cases of clearest improvement—Realistic and Fatigue—share a common structure: the teacher provides reliable feedback for a portion of training, then becomes progressively less reliable through combined noise sources or temporal degradation. The BRF’s running trust estimate allows it to capitalize on early reliable signal while attenuating later degraded feedback. The Naive baseline, which assigns full weight to all feedback, cannot distinguish reliable steps from unreliable ones and is misled in proportion to the noise.

The trust weight evolution plots confirm this mechanistically: the filter converges to accurate reliability estimates within 2,000–5,000 steps for most teacher types, well within the 15,000-step training budget.

5.2 Recoverable vs. Unrecoverable Noise

Both PPO and DQN achieve 0.000 success under StochasticHigh and Adversarial teachers, yet StochasticLow yields nonzero success for both agents in the BRF condition. This pattern suggests the existence of a *recoverable noise threshold* somewhere between 10% and 80% flip probability. Below this threshold, the BRF can extract enough directional signal to improve on the sparse baseline. Above it, the feedback is too corrupted to provide a useful learning signal within the training budget.

Importantly, the Adversarial trust curve does collapse correctly to near zero—the filter correctly identifies the teacher as unreliable. The failure is not a misidentification; it is a constraint of the reward structure: with a terminal reward of +1.0 occurring rarely in 15,000 steps and human feedback contributing at most ± 0.1 per step, there is insufficient environmental signal to compensate for universally corrupt feedback.

5.3 Environment Complexity and Training Budget

LavaGap and DoorKey saw minimal or zero success across all methods and agents. Since all three reward modes perform similarly on these environments, the bottleneck is the training budget rather than reward signal quality. DoorKey’s three-stage sequential structure requires the agent to discover multiple sparse sub-goals in sequence—a substantially harder credit-assignment problem than the single-stage Empty environment. These results indicate that the BRF’s contribution is most measurable in settings where the base task is solvable within the training budget.

The absence of trust decay in the Fatigue curves (Section 4.1) is also explained by the training budget: the Fatigue teacher’s noise is designed to peak at 50,000 steps. At 15,000 steps the teacher is still in its low-noise phase, and the trust weight correctly reflects high reliability at that point in training.

5.4 DQN and the Off-Policy Limitation

DQN’s weaker results relative to PPO arise in part from a structural incompatibility between the BRF and experience replay. DQN stores past transitions in a replay buffer and samples from them uniformly during training. Transitions collected early—before the trust estimate converged—carry a modified reward computed using an inaccurate early trust weight. Once stored, this reward is fixed in the buffer. Even after the filter has learned that the teacher is unreliable, those corrupted early transitions continue to influence policy updates when replayed.

PPO avoids this issue because it is on-policy: every gradient update uses only freshly collected transitions, where the current trust weight is applied. The BRF’s per-step correction is therefore immediately effective for PPO but only partially effective for DQN.

5.5 Limitations

- **Limited seeds.** Experiments use 3 random seeds, producing high variance in per-condition estimates. Results should be interpreted as directional rather than statistically definitive.
- **Short training budget.** At 15,000 timesteps, the Fatigue teacher’s noise never reaches its intended maximum, and DoorKey and LavaGap environments do not converge under any

condition. Conclusions about BRF performance in these settings are necessarily limited.

- **Single environment for key findings.** The most meaningful BRF results are concentrated in MiniGrid-Empty-8x8-v0. Broader validation across more complex solvable environments would strengthen the generalizability of the findings.